

AIRLINERS

THE INTRODUCTION OF THE BOEING 747 in 1970 marked the start of the contemporary era in air travel. The fiercely competitive nature of the airliner market since then has led to the fall of several great names. Lockheed pulled out of airliner production after unsuccessfully running the L-1011 TriStar against the DC-10, and Douglas itself became McDonnell Douglas, before being absorbed into Boeing. In Europe, Airbus Industrie emerged as a serious rival to Boeing, successfully challenging the American company's dominance of the world market. Once it became clear that the future did not lie with supersonic travel, the aircraft themselves were inevitably committed to a process of evolution rather than revolution. Jet engines became progressively more powerful, but also more fuel-efficient and quieter.

Avionics improved rapidly, with fly-by-wire controls and improvements in all forms of electronic equipment.

AIR FORCE ONE

Since 1990, the US president's official aircraft have been modified variants of the Boeing 747. See pages 394–95.



Airbus A320-200



Aimed at the 150-seat market, the advanced A320 first flew in February 1987. With its fully computerized cockpit and fly-by-wire control system, the design set the standard by which all future airliners were judged. Based on the A320, Airbus developed a further two jets: the A319 (up to 120 passengers) and the A321 (up to 220 passengers). All three remain in production.

ENGINE 2 x 25,000lbs (11,350kg) CFM56-5A1 turbofan

WINGSPAN 111ft 3in (33.9m) **LENGTH** 123ft 3in (37.6m)

CRUISING SPEED 550mph (903kph)

PASSENGERS 150–179 **CREW** 2

Airbus A330-200

The combined A330/A340 program was launched in June 1987, with both designs sharing the same wings, airframe, cockpit, and systems. Only the number and type of engines differed. The first to fly was the long-range, four-engined A340 in October 1991, followed by the medium-range, twin-engined A330 in November 1992. Airbus now offers many permutations of passenger capacity and range.



ENGINE 2 x 68,000lbs (30,870kg) Rolls-Royce Trent 768

WINGSPAN 197ft 10in (60.3m) **LENGTH** 193ft 7in (59m)

CRUISING SPEED 547mph (880kph)

PASSENGERS 256–293 **CREW** 2

Boeing 737-700



Boeing offered its "Next Generation" family of medium-range 737s (737-600 onward) with a new wing permitting a higher cruise speed and more efficient CFM International engines giving longer range. The 737-700 and -800 first flew in 1997, and the -900 in 2000. By 2016 Boeing had delivered more than 9,000 737s of all types, with more than 4,000 on order.

ENGINE 2 x 24,200lbs (10,990kg) CFM56-7B turbofan

WINGSPAN 112ft 7in (34.3m) **LENGTH** 110ft 4in (33.6m)

CRUISING SPEED 530mph (853kph)

PASSENGERS 126–149 **CREW** 2

Boeing 757-300

The narrow-body (single-aisle) 757 was developed in parallel with the wide-body 767. Conceived as a replacement for the 727, the first 757 flew in 1979, a few months after the initial 767. Offered with a choice of engines from either Pratt & Whitney or Rolls-Royce, sales were rather slow at first but improved into the late 1980s, as the Rolls-Royce version became the preferred choice. Production ended in 2005, with 1,050 built.



ENGINE 2 x 43,500lbs (19,750kg) R-R RB211-535E4B turbofan

WINGSPAN 124ft 10in (38.1m) **LENGTH** 178ft 7in (54.5m)

CRUISING SPEED 530mph (853kph)

PASSENGERS 240–289 **CREW** 2

Boeing 767

The 767 was developed as a medium-to-long-range, wide-body (twin-aisle) airliner in parallel with its narrow-body sister, the 757, and first took to the skies in September 1981. Sharing an identical cockpit to the 757, a pilot trained on one could fly the



ENGINE 2 x 63,500lbs (28,830kg) GE CF6-80C turbofan

WINGSPAN 170ft 4in (51.9m) **LENGTH** 201ft 4in (61.4m)

CRUISING SPEED 530mph (853kph)

PASSENGERS 245–375 **CREW** 2

28,830kgp General Electric turbofan engine

other without additional training, offering considerable savings to airlines buying both. The largest of the family is the stretched -400ER (Extended Range) which entered service in September 2000. In 2016, total deliveries of 767s reached 1,000. The aircraft remains in production despite the advent of the 787 Dreamliner.